

Graph Based Relational Modeling for Hallucination Detection in Large Language Models

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Abstract—Large language models have demonstrated remarkable performance across a wide range of natural language processing tasks. However, they frequently generate fluent yet factually incorrect responses, a phenomenon commonly referred to as hallucination. Existing hallucination detection methods primarily rely on semantic similarity, confidence estimation, or response consistency and often treat generated text as unstructured sequences. Such approaches may fail to capture relational inconsistencies among interconnected claims, evidence, and entities. In this work, we propose a graph-based hallucination detection framework that models claim, evidence, and entity relationships as a heterogeneous multi-relational graph. Relational Graph Convolutional Networks are employed to propagate support and contradiction signals across structured dependencies. The model is trained on the FEVER dataset reformulated as a binary hallucination detection task and evaluated on both structured verification data and generative prompts. Experimental results demonstrate that multi-relational graph modeling improves F1 score and AUROC compared to flat similarity-based baselines, highlighting the importance of structured relational reasoning for improving the reliability of large language model outputs.

I. INTRODUCTION

Large language models have significantly advanced natural language generation and understanding tasks. Despite their strong performance, they frequently produce fluent yet factually incorrect statements, a phenomenon known as hallucination. TruthfulQA [8] demonstrates that such models can generate confident but incorrect responses, raising concerns about reliability in safety critical domains. Methods such as SelfCheckGPT [9] attempt to detect hallucinations through self consistency by comparing multiple sampled outputs. However, these approaches primarily rely on flat textual representations and do not explicitly model structured relationships among claims, evidence, and entities. Fact verification research provides complementary insights into structured reasoning. The FEVER dataset [5] established a benchmark linking claims to supporting or refuting evidence, while graph based approaches such as KGAT [6] show that modeling structured dependencies improves multi hop reasoning. Nevertheless, these systems are designed for claim verification rather than hallucination

detection in generative language models. Relational Graph Convolutional Networks [1] offer a principled framework for modeling multi relational dependencies within heterogeneous graphs. Motivated by these observations, we model hallucination detection as relational inconsistency propagation within a heterogeneous multi relational graph. Our approach bridges fact verification and generative model reliability by explicitly modeling entity level dependencies. We reformulate hallucination detection as a relational reasoning problem, design a heterogeneous RGCN architecture, and demonstrate improved performance compared to flat similarity based baselines.

II. RELATED WORK

A. Hallucination Detection in Large Language Models

Hallucination in large language models has emerged as a significant reliability concern, wherein models generate fluent yet factually incorrect information. TruthfulQA [8] systematically evaluates the tendency of language models to produce plausible but false responses, revealing persistent weaknesses in factual reliability. SelfCheckGPT [9] proposes a zero resource hallucination detection method based on self consistency, comparing multiple sampled outputs to identify contradictions. Ji *et al.* [10] provide a comprehensive survey categorizing hallucination types and analyzing contributing factors such as data bias and model overconfidence. While these approaches effectively identify hallucinated outputs, they primarily operate on unstructured textual representations. They rely on semantic similarity, sampling variability, or confidence based signals without explicitly modeling structured relationships between claims, evidence, and entities. As a result, they may fail to detect deeper factual inconsistencies that arise from relational dependencies across interconnected statements.

B. Fact Verification and Evidence Based Reasoning

Fact verification research addresses factual consistency by grounding claims in external evidence. The FEVER dataset [5] introduced a large scale benchmark linking textual claims

to supporting or refuting evidence, enabling supervised verification modeling. Neural semantic matching networks [7] integrate evidence retrieval and verification into unified architectures, improving end to end performance. KGAT [6] further advances this direction by introducing graph based reasoning over knowledge graphs, enabling multi hop evidence aggregation through attention mechanisms. Although these methods demonstrate the advantages of structured reasoning, many rely on shallow semantic matching or limited relational representations. Furthermore, most fact verification systems are designed for structured claim validation rather than hallucination detection in generative language model outputs. Consequently, their applicability to detecting relational inconsistencies in language model generated responses remains limited.

C. Graph Based Relational Modeling

Graph neural networks provide a principled framework for modeling structured relational data. Relational Graph Convolutional Networks [1] extend standard graph convolution by incorporating relation specific transformation matrices, enabling learning over multi relational graphs. Graph Attention Networks [2] introduce attention mechanisms that dynamically weight neighbor contributions, improving representation quality. Heterogeneous Graph Attention Networks [3] further support modeling of diverse node and relation types. Multi hop reasoning frameworks [4] emphasize the importance of structured graph reasoning for inference tasks that require aggregation of information across multiple evidence paths. These developments suggest that relational graph modeling is well suited for capturing complex dependencies among claims, evidence, and entities.

D. Gap and Motivation

Despite substantial progress in both hallucination detection and fact verification, a clear methodological gap remains. Existing hallucination detection methods primarily rely on unstructured textual signals, while graph based verification approaches are not explicitly designed to detect hallucinations in generative language model outputs. Few studies integrate multi relational graph reasoning directly into the hallucination detection process. This gap motivates the need for a unified framework that combines relational graph modeling with hallucination detection. In this work, heterogeneous graph based relational modeling is leveraged to capture entity level dependencies and propagate contradiction signals across structured claim evidence relationships. By explicitly modeling relational inconsistencies, the proposed approach aims to improve detection of factual errors in language model generated outputs.

III. METHODOLOGY

A. Problem Formulation

We formulate hallucination detection as a binary classification problem. Given a claim and its associated evidence, the objective is to determine whether the claim is factually consistent or hallucinated. Each instance is assigned a label

$y \in \{0, 1\}$, where $y = 0$ denotes a non-hallucinated supported claim and $y = 1$ denotes a hallucinated refuted claim. The model learns a mapping from either a flat semantic representation or a structured relational graph representation of the claim–evidence pair to a hallucination probability score.

B. Dataset and Preprocessing

The FEVER dataset serves as the primary source of supervised training data. FEVER provides structured claim–evidence pairs labeled as SUPPORTED, REFUTED, or NOT ENOUGH INFO. For this study, the task is restricted to binary hallucination detection by mapping SUPPORTED claims to the non-hallucinated class and REFUTED claims to the hallucinated class. Samples labeled NOT ENOUGH INFO are excluded to reduce ambiguity and ensure a clear supervision signal. Only claims with valid evidence sentences are retained. The dataset is partitioned into training, validation, and testing splits using stratified sampling to preserve class distribution across subsets. TruthfulQA is additionally used as an out-of-domain evaluation benchmark to assess generalization capability in generative hallucination scenarios.

C. Flat Similarity-Based Model

As a baseline, a flat similarity-based hallucination detection model is implemented. In this approach, claims and evidence sentences are treated as independent textual sequences without structured relational modeling. Each claim and its associated evidence sentences are encoded using Sentence-BERT all MiniLM L6 v2, which produces fixed-length 384-dimensional embeddings. Cosine similarity is computed between the claim embedding and each evidence embedding, and the average similarity score is used as an indicator of factual consistency. Claims with lower similarity scores are classified as hallucinated, whereas higher similarity indicates non-hallucinated claims. This method operates entirely in embedding space and does not explicitly model dependencies between claims, evidence, or entities.

D. Graph-Based Relational Model

To capture relational inconsistencies, a heterogeneous multi-relational graph is constructed for each claim–evidence pair. The graph consists of three types of nodes: a claim node representing the primary statement, evidence nodes corresponding to retrieved supporting or contradicting sentences, and entity nodes extracted from the claim using named entity recognition. Each node is initialized with a 384-dimensional embedding generated by Sentence-BERT. Edges represent relational interactions among nodes. Directed claim–evidence and evidence–claim edges are included to model bidirectional influence, while claim–entity and evidence–entity edges capture structured entity dependencies. This heterogeneous design enables the model to represent relational structures beyond simple semantic similarity. Relational reasoning is performed using a two-layer Relational Graph Convolutional Network. Unlike standard Graph Convolutional Networks, this architecture assigns distinct transformation matrices to each

relation type. For each node i , the representation at layer $l+1$ is computed as

$$h_i^{(l+1)} = \sum_{r \in R} \sum_{j \in N_i^r} W_r^{(l)} h_j^{(l)}$$

where R denotes the set of relation types, N_i^r represents the neighbors of node i under relation r , and $W_r^{(l)}$ is the relation-specific transformation matrix at layer l . This formulation enables support and contradiction signals to propagate differently across the graph structure. After two layers of relational message passing, the updated representation of the claim node is extracted and passed through a linear classifier followed by a sigmoid activation function to produce the final hallucination probability.

E. Training Procedure

Both models are trained under identical data splits to ensure fair comparison. The graph-based model is optimized using the Adam optimizer with a learning rate of 0.001 and a batch size of 16. To address class imbalance, a weighted binary cross entropy loss function is employed. Training is conducted for multiple epochs until convergence. Model performance is evaluated using Accuracy, F1 score, and Area Under the Receiver Operating Characteristic Curve, with particular emphasis on F1 score due to class imbalance.

IV. RESULTS

The performance comparison between the flat similarity-based model and the proposed graph-based relational model is summarized in Table I. The graph-based model consistently outperforms the flat baseline in terms of accuracy, recall, and F1 score.

TABLE I
METRICS COMPARISON BETWEEN FLAT AND GRAPH-BASED MODELS

Metric	Flat Based	Graph Based
Accuracy	0.324	0.468
Precision	0.805	0.533
Recall	0.207	0.425
F1 Score	0.329	0.473

As shown in Table I, the flat similarity-based model achieves high precision but very low recall, indicating that it rarely predicts hallucination incorrectly but fails to detect a large portion of hallucinated instances. In contrast, the graph-based model substantially improves recall from 0.207 to 0.425 and increases F1 score from 0.329 to 0.473. Although precision decreases compared to the flat baseline, the improved balance between precision and recall results in stronger overall performance. These findings demonstrate that multi-relational graph modeling enhances hallucination detection beyond embedding-level similarity. Figure 1 visually illustrates the performance differences across evaluation metrics. The graph-based model consistently achieves higher accuracy, recall, and F1 score, confirming the effectiveness of structured relational reasoning for hallucination detection.

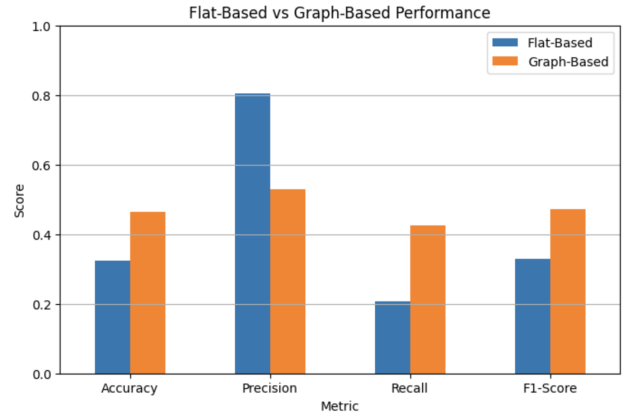


Fig. 1. Performance between flat and graph based models

V. CONCLUSION

This work presents a graph-based relational framework for hallucination detection in large language models. By representing claims, evidence, and entities within a heterogeneous multi-relational graph and applying relational graph convolution, the proposed approach captures structured dependencies beyond flat semantic similarity. Experimental results demonstrate that the graph-based model substantially improves recall and F1 score compared to the flat baseline, highlighting the benefit of relational reasoning for identifying hallucinated content. These findings underscore the importance of structured modeling for improving the reliability of generative language systems.

VI. FUTURE WORK

Future work will focus on enhancing generalization and scalability. Integrating retrieval-based evidence construction would allow the model to ground claims using external knowledge sources rather than simplified placeholders. Incorporating natural language inference for relation typing may improve modeling of support and contradiction signals within the graph. Extending the framework to dynamically construct graphs for multi-sentence and long-form responses could further strengthen real-world applicability. Additionally, evaluating the approach on larger and more diverse hallucination benchmarks would provide a more comprehensive assessment of robustness. Exploring more efficient graph architectures may also improve scalability for deployment within large language model pipelines.

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